Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Momentum & Impulse

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A particle of mass 2 kg moving at $5\,\mathrm{m\,s^{-1}}$ is brought to rest by an impulse. Find the magnitude of the impulse. **[3]**

2. A particle of mass 3 kg has its velocity changed from $4\,\mathrm{m\,s^{-1}}$ to $10\,\mathrm{m\,s^{-1}}$ by an impulse. Find the impulse. **[3]**

3. A particle of mass 5 kg moving at $6\,\mathrm{m\,s^{-1}}$ receives an impulse of $15\,\mathrm{N\,s}$ in the direction of motion. Find its new speed. **[3]**

4. A particle of mass 4 kg moving at $3\,\mathrm{m\,s^{-1}}$ is struck by an impulse of $8\,\mathrm{N\,s}$ opposite to its motion. Find its new velocity. **[3]**

5. Two particles of masses 2 kg and 3 kg moving towards each other at $4\,\mathrm{m\,s^{-1}}$ and $2\,\mathrm{m\,s^{-1}}$ respectively collide and coalesce. Find the common speed after impact. **[4]**

6. Two particles of masses 5 kg and 3 kg moving in the same direction at $6\,\mathrm{m\,s^{-1}}$ and $2\,\mathrm{m\,s^{-1}}$ collide and stick together. Find the common speed. **[4]**

7. A particle of mass 2 kg moving at $5\,\mathrm{m\,s^{-1}}$ collides with a stationary particle of mass 3 kg. After impact they move together. Find the common speed and the impulse. **[5]**

8. A ball of mass 0.5 kg hits a wall at $8\,\mathrm{m\,s^{-1}}$ and rebounds at $6\,\mathrm{m\,s^{-1}}$. Find the impulse exerted by the wall. **[3]**

9. A particle of mass m moving at speed u is brought to rest. Find the impulse. **[2]**

10. Two particles of equal mass m approach each other at equal speeds u and collide head-on. After impact they move in opposite directions each at speed $\frac{1}{2}u$. Find the impulse on each. **[4]**

11. A particle of mass 4 kg has momentum $12\,\text{kg}\,\text{m}\,\text{s}^{-1}$. Find its speed. **[2]**

12. An impulse of 10 N s acts on a particle of mass 2 kg initially at rest. Find its final speed. **[2]**

13. A particle of mass 3 kg moving at $5\,\text{m}\,\text{s}^{-1}$ collides with a stationary particle of mass 2 kg. After impact the 3 kg particle has speed $1\,\text{m}\,\text{s}^{-1}$ in the same direction. Find the speed of the 2 kg particle. **[4]**

14. A particle of mass 5 kg moving at $4\,\text{m}\,\text{s}^{-1}$ receives an impulse of $10\mathbf{i} + 5\mathbf{j}$ N s. Find its new velocity if its original velocity was $4\mathbf{i}\,\text{m}\,\text{s}^{-1}$. **[4]**

15. Two particles collide and the total momentum after impact equals the total momentum before impact. What principle is this? **[2]**

Section B – Hard (Q16–25)

16. Particles of masses 2 kg and 4 kg moving in opposite directions at $5\,\text{m}\,\text{s}^{-1}$ and $3\,\text{m}\,\text{s}^{-1}$ collide. After impact the 2 kg particle is at rest. Find the speed of the 4 kg particle and the impulse. **[5]**

17. A particle of mass 3 kg moving at $6\,\text{m}\,\text{s}^{-1}$ collides with a stationary particle of mass 6 kg. The coefficient of restitution is 0.5. Find the speeds after impact. **[6]**

18. Two particles of masses m and $2m$ approach each other with speeds $2u$ and u. The coefficient of restitution is e. Find the speeds after impact in terms of u and e. **[6]**

19. A sphere of mass 2 kg moving at $4\,\text{m}\,\text{s}^{-1}$ hits a stationary sphere of mass 3 kg. After impact the first sphere continues at $1\,\text{m}\,\text{s}^{-1}$ in the same direction. Find the speed of the second sphere and the coefficient of restitution. **[6]**

20. A particle of mass 4 kg moving at $5\,\mathrm{m\,s^{-1}}$ collides directly with a particle of mass 6 kg moving in the same direction at $2\,\mathrm{m\,s^{-1}}$. The coefficient of restitution is 0.4. Find the speeds after impact. **[6]**

21. A ball hits a smooth vertical wall at $10\,\mathrm{m\,s^{-1}}$ at 30° to the normal. The coefficient of restitution is 0.6. Find the speed and direction after impact. **[6]**

22. Two particles of masses 5 kg and 3 kg are moving towards each other at $4\,\mathrm{m\,s^{-1}}$ and $6\,\mathrm{m\,s^{-1}}$. After impact they move in opposite directions at $2\,\mathrm{m\,s^{-1}}$ and v. Find v and the impulse. **[5]**

23. A particle of mass 2 kg is moving with velocity $3\mathbf{i} - 2\mathbf{j}$. It receives an impulse $4\mathbf{i} + 6\mathbf{j}$. Find its new velocity. **[4]**

24. Show that if two particles of equal mass collide directly and the coefficient of restitution is e, the relative speed of separation is e times the relative speed of approach. **[4]**

25. A bullet of mass 0.02 kg is fired at $400\,\mathrm{m\,s^{-1}}$ into a stationary block of mass 2 kg which can move freely. The bullet embeds. Find the common speed and the kinetic energy lost. **[6]**

End of Worksheet • *Show all working clearly.*